

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Easy

Question

$$2\ell + 2w \leq 27$$

A rectangle has length ℓ and width w . The inequality gives the possible values of ℓ and w for which the perimeter of this rectangle is less than or equal to **27**. Which statement is the best interpretation of $(\ell, w) = (8, 3)$ in this context?

Answer

- A. If the rectangle has length **3** and width **8**, its perimeter is less than or equal to **27**.
- B. If the rectangle has length **8** and width **3**, its perimeter is less than or equal to **27**.
- C. If the rectangle has length **3** and width **8**, its perimeter is greater than or equal to **27**.
- D. If the rectangle has length **8** and width **3**, its perimeter is greater than or equal to **27**.

Correct Answer: B

Rationale

Choice B is correct. It’s given that a rectangle has length ℓ and width w , and the inequality $2\ell + 2w \leq 27$ gives the possible values of ℓ and w for which the perimeter of this rectangle is less than or equal to **27**. To determine the best interpretation of $(\ell, w) = (8, 3)$ in this context, the values can be substituted in the given inequality. Substituting **8** for ℓ and **3** for w in this inequality yields $2(8) + 2(3) \leq 27$, which is equivalent to $16 + 6 \leq 27$, or $22 \leq 27$. Since this inequality is true, if the rectangle has length **8** and width **3**, its perimeter is less than or equal to **27**.

Choice A is incorrect. The interpretation of $(\ell, w) = (8, 3)$ implies that the rectangle has length **8** and width **3**, not length **3** and width **8**.

Choice C is incorrect. The interpretation of $(\ell, w) = (8, 3)$ implies that the rectangle has length **8** and width **3**, not length **3** and width **8**.

Choice D is incorrect and may result from conceptual or calculation errors.